

How to prove that certain languages are not regular?

Pumping lemma

- Many languages are not regular.
- **Pumping lemma** is a method to prove that certain languages are not regular.

Pumping property

- If a language is regular, then it must have the **pumping property**.
- If a language does not have the pumping property, then the language is not regular. ▷ Proof by contraposition

How to prove languages non-regular using pumping lemma?

- **Proof by contradiction.**

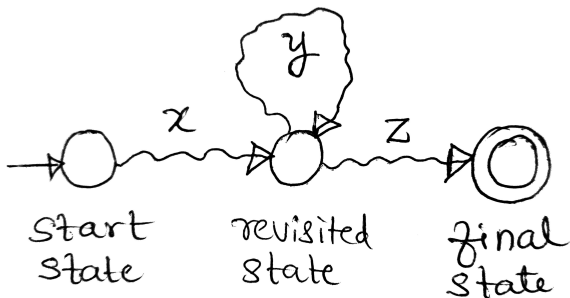
Assume that the language is regular.

Show that the language does not have the pumping property.

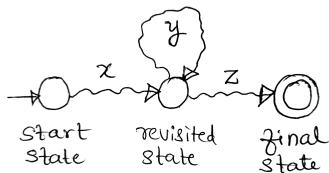
Contradiction! Hence, the language has to be non-regular.

Pumping property of regular languages

- Suppose a DFA M with s number of states accepts a very long string w such that $|w| \geq s$ from a language L .
- From **pigeonhole principle**, at least one state is visited twice.
- This implies that the string went through a **loop**.



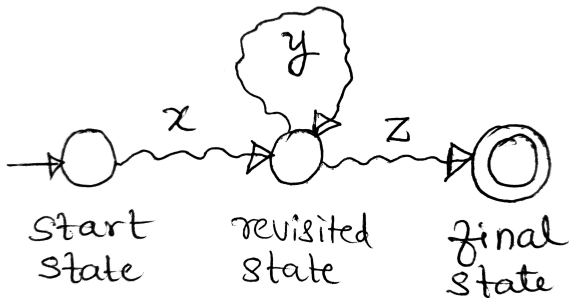
Pumping property of regular languages



Observations

- Suppose string w has more characters than the number of states in the DFA, i.e., $|w| \geq s$
- String w can be split into three parts i.e., $w = xyz$ where
 - x : string before the first loop
 - y : string of the first loop
 - z : string after the first loop (might contain loops)
- Loop must appear i.e., $|y| \geq 1$
(x and z can be empty)
- Loop must appear in the first s characters of w i.e., $|xy| \leq s$

Pumping property of regular languages



Idea

- An infinite number of strings can be pumped with loop length and they must also be in the language.
- Formally, for all $i \geq 0$, xy^iz must be in the language.
- xz , xyz , $xyyz$, $xyyyz$, etc must also belong to the language.

Pumping lemma for regular languages

Theorem

Suppose L is a regular language over alphabet Σ . Suppose L is accepted by a finite automaton M having s states. Then, every long string $w \in L$ satisfying $|w| \geq s$ can be split into three strings $w = xyz$ such that the following three conditions are true.

- $|xy| \leq s$.
- $|y| \geq 1$.
- For every $i \geq 0$, the string xy^iz also belongs to L .

$L = \{a^n b^n \mid n \geq 0\}$ is non-regular

Problem

- Prove that $L = \{a^n b^n \mid n \geq 0\}$ is not a regular language.

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Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $w = a^s b^s$.
- Let $w = xyz = \boxed{a^p} \boxed{a^q} \boxed{a^r b^s}$
where $|xy| \leq s$, $|y| \geq 1$, and $p + q + r = s$.
- Also, $xy^i z$ must belong to L for all $i \geq 0$.
- But, $xyyz$ is not in L .
Reason: $xyyz = a^p a^q a^q a^r b^s = a^{s+q} b^s \notin L$.
 $xyyz$ has more a 's than b 's.
- Contradiction! Hence, L is not regular.

$L = \{w \mid n_a(w) = n_b(w)\}$ is non-regular

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Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $w = (ab)^s$.
- Let $w = xyz = \boxed{\epsilon} \boxed{(ab)^1} \boxed{(ab)^{s-1}}$
- We have $|xy| \leq s$ and $|y| \geq 1$.
- Also, xy^iz must belong to L for all $i \geq 0$.
- xy^iz belongs to L for all $i \geq 0$.
- No contradiction! Hence, L is regular.

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- We have $|xy| \leq s$ and $|y| \geq 1$.
- Also, xy^iz must belong to L for all $i \geq 0$.
- xy^iz belongs to L for all $i \geq 0$.
- No contradiction! Hence, L is regular.

Mistakes

Incorrect solution! Why? Multiple reasons:

1. If we cannot find a contradiction, that does not prove anything.
2. We must try for all possible values of x, y such that $|xy| \leq s$.
3. The chosen string $(ab)^s$ is a bad string to work on.

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- Suppose $w = a^s b^s$.
- Let $w = xyz = \boxed{a^p} \boxed{a^q} \boxed{a^r b^s}$
where $|xy| \leq s$, $|y| \geq 1$, and $p + q + r = s$.
- Also, $xy^i z$ must belong to L for all $i \geq 0$.
- But, $xyyz$ is not in L .
Reason: $xyyz = a^p a^q a^q a^r b^s = a^{s+q} b^s \notin L$.
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where $|xy| \leq s$, $|y| \geq 1$, and $p + q + r = s$.

- Also, $xy^i z$ must belong to L for all $i \geq 0$.
- But, $xyyz$ is not in L .

Reason: $xyyz = a^p a^q a^q a^r b^s = a^{s+q} b^s \notin L$.

$xyyz$ has more a 's than b 's.

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Takeaway

1. **Reduction!** Reduce a problem to another. Reuse its solution.

Superset of a non-regular language

Problem

- $\{a^n b^n\}$ is a subset of $\{w \mid n_a(w) = n_b(w)\}$.

We used the fact that $\{a^n b^n\}$ is non-regular to prove that $\{w \mid n_a(w) = n_b(w)\}$ is non-regular.

Is a superset of a non-regular language non-regular?

Superset of a non-regular language

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- $\{a^n b^n\}$ is a subset of $\{w \mid n_a(w) = n_b(w)\}$.

We used the fact that $\{a^n b^n\}$ is non-regular to prove that $\{w \mid n_a(w) = n_b(w)\}$ is non-regular.

Is a superset of a non-regular language non-regular?

Solution

- No!

Σ^* is a superset of every non-regular language.

But, it is regular.

$L = \{w \mid n_a(w) = n_b(w)\}$ is non-regular

Problem

- Prove that $L = \{w \mid n_a(w) = n_b(w)\}$ is not a regular language.

Solution (without using pumping lemma)

- Suppose L is regular.
- We know that $L' = \{w \mid w = a^i b^j, i, j \geq 0\}$ is regular.
- As regular languages are closed under intersection, $L \cap L'$ must also be regular.
- We see that $L \cap L' = \{w \mid w = a^n b^n \text{ and } n \geq 0\}$.
- But, this language was earlier proved to be non-regular.
- Contradiction! Hence, L is not regular.

$L = \{ww\}$ is non-regular

Problem

- Prove that $L = \{ww\}$ is not a regular language.

$L = \{ww\}$ is non-regular

Problem

- Prove that $L = \{ww\}$ is not a regular language.

Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $ww = a^s a^s$.
- Let $ww = xyz = \boxed{a^p} \boxed{a^1} \boxed{a^{s-p-1} a^s}$
- We have $|xy| \leq s$ and $|y| \geq 1$.
- Also, $xy^i z$ must belong to L for all $i \geq 0$.
- But, $xyyz$ is not in L .

Reason: $xyyz = a^p a^1 a^1 a^{s-p-1} a^p = a^{s+1} a^s \notin L$.

$xyyz$ has odd number of a 's.

- Contradiction! Hence, L is not regular.

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- Prove that $L = \{ww\}$ is not a regular language.

Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $ww = a^s a^s$.
- Let $ww = xyz = \boxed{a^p} \boxed{a^1} \boxed{a^{s-p-1}a^s}$
- We have $|xy| \leq s$ and $|y| \geq 1$.
- Also, $xy^i z$ must belong to L for all $i \geq 0$.
- But, $xyyz$ is not in L .
Reason: $xyyz = a^p a^1 a^1 a^{s-p-1} a^p = a^{s+1} a^s \notin L$.
 $xyyz$ has odd number of a 's.
- Contradiction! Hence, L is not regular.

Mistakes

Incorrect solution! Why?

1. We must try all possible values of x, y such that $|xy| \leq s$.
2. Try pumping with $y \in \{a^2, a^4, \dots\}$ such that $|y| \leq s$.

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Problem

- Prove that $L = \{ww\}$ is not a regular language.

Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $ww = a^s b^s a^s b^s$.
- Let $ww = xyz = \boxed{a^p} \boxed{a^q} \boxed{a^r b^s a^s b^s}$
where $|xy| \leq s$, $|y| \geq 1$, and $p + q + r = s$.
- Also, $xy^i z$ must belong to L for all $i \geq 0$.
- But, $xyyz$ is not in L .
Reason: $xyyz = a^p a^q a^q a^r b^s a^s b^s = a^{s+q} b^s a^s b^s \notin L$.
- Contradiction! Hence, L is not regular.

$L = \{w \mid w = a^n, n \text{ is a square}\}$ is non-regular

Problem

- Prove that $L = \{w \mid w = a^{n^2}\}$ is not a regular language.

$L = \{w \mid w = a^n, n \text{ is a square}\}$ is non-regular

Problem

- Prove that $L = \{w \mid w = a^{n^2}\}$ is not a regular language.

Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $w = a^{s^2}$.
- Let $w = xyz = \boxed{a^p} \boxed{a^q} \boxed{a^r a^{s^2-s}}$
where $|xy| \leq s$, $|y| \geq 1$, and $p + q + r = s$.
- Also, xy^iz must belong to L for all $i \geq 0$.
- But, $xyyz$ is not in L .
Reason: $xyyz = a^p a^q a^q a^r a^{s^2-s} = a^{s^2+q} \notin L$.
Because, $s^2 < s^2 + q < (s+1)^2$.
- Contradiction! Hence, L is not regular.

$L = \{w \mid w = a^n, n \text{ is prime}\}$ is non-regular

Problem

- Prove that $L = \{w \mid w = a^n, n \text{ is prime}\}$ is not regular.

$L = \{w \mid w = a^n, n \text{ is prime}\}$ is non-regular

Problem

- Prove that $L = \{w \mid w = a^n, n \text{ is prime}\}$ is not regular.

Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $w = a^m$, where m is prime and $m \geq s$.
- Let $w = xyz = \boxed{a^p} \boxed{a^q} \boxed{a^r}$
where $|xy| \leq s$, $|y| \geq 1$, and $p + q + r = m$.
- Also, xy^iz must belong to L for all $i \geq 0$.
- But, $xy^{m+1}z$ is not in L .
Reason: $xy^{m+1}z = a^p a^{q(m+1)} a^r = a^{m(q+1)} \notin L$.
- Contradiction! Hence, L is not regular.

$L = \{w \mid w = a^m b^n, m > n\}$ is non-regular

Problem

- Prove that $L = \{w \mid w = a^m b^n, m > n\}$ is not regular.

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Problem

- Prove that $L = \{w \mid w = a^m b^n, m > n\}$ is not regular.

Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $w = a^{s+1}b^s$.
- Let $w = xyz = \boxed{a^p} \boxed{a^q} \boxed{a^r b^s}$
where $|xy| \leq s$, $|y| \geq 1$, and $p + q + r = s + 1$.
- Also, $xy^i z$ must belong to L for all $i \geq 0$.
- But, xz is not in L . ▷ Pumping down
Reason: $xz = a^p a^r b^s = a^{p+r} b^s \notin L$.
Because, $p + r \leq s$ i.e., $\#a$'s is not greater than $\#b$'s.
- Contradiction! Hence, L is not regular.

$L = \{w \mid w = a^m b^n, m \neq n\}$ is non-regular

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Problem

- Prove that $L = \{w \mid w = a^m b^n, m \neq n\}$ is not regular.

Solution

- Suppose L is regular. Then it must satisfy pumping property.
- Suppose $w = a^s b^{s+s!}$.
- Let $w = xyz = \boxed{a^p} \boxed{a^q} \boxed{a^r b^{s+s!}}$
where $|xy| \leq s$, $|y| \geq 1$, and $p + q + r = s$.
- Also, $xy^i z$ must belong to L for all $i \geq 0$.
- But, $xy^i z$ is not in L for some i .
We pump a^q to get $a^{s+s!} b^{s+s!}$.
Reason: $xy^i z = a^p a^{qi} a^r b^{s+s!} = a^{s+(i-1)q} b^{s+s!} \notin L$.
This means $(i-1)q = s! \implies i = s!/q + 1$.
- Contradiction! Hence, L is not regular.

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Problem

- Prove that $L = \{w \mid w = a^m b^n, m \neq n\}$ is not regular.

Solution (without using pumping lemma)

- Suppose L is regular.
- We know that $L' = \{w \mid w = a^i b^j, i, j \geq 0\}$ is regular.
- Let $L'' = \{w \mid w = a^n b^n, n \geq 0\}$.
- As regular languages are closed under intersection and complementation, $L = L' - L'' = L' \cap \bar{L''}$ is regular.
This implies that L'' is regular.
- But, the language L'' was earlier proved to be non-regular.
- Contradiction! Hence, L is not regular.